

Problem 10

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Problem 1 For the piecewise function f defined by

$$f(x) = \begin{cases} \sin(x) & \text{if } x \leq 0 \\ \frac{x^2}{x^2 - 4} & \text{if } 0 < x < 2 \\ \frac{x^2}{x^2 + 4} & \text{if } 2 \leq x \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.